

Leo Torres

Senior Data Engineer

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Technical Skills

Languages & Frameworks

- Python (Expert): pandas, NumPy, SciPy, FastAPI, NetworkX, scikit-learn
- JavaScript (Advanced): Vue.js, HTMX, Node.js, D3.js
- AI & LLMs: GPT-4, Claude, RAG systems, vector databases, prompt engineering
- Other: SQL (advanced), Mathematica, C/C++, R, Julia

Infrastructure & Tools

- Cloud & Data: GCP (BigQuery, Dataflow, Cloud Run), Azure ML Studio
- Databases: BigQuery, PostgreSQL, experience with billion-row datasets
- DevOps: Docker, Airflow, GitHub Actions, Netlify, Fly.io
- Development: Git, pytest, CI/CD, OOP design, performance optimization

Experience

FGS Global - Lead Data Scientist & Tech Lead
Berlin, Germany

May 2023 - Present

- Built LLM-powered pipelines processing 1M+ news articles daily for Fortune 500 clients
- Architected data infrastructure handling 1B+ log entries with sub-second query performance
- Technical lead for a team of 6 engineers on the flagship internal product
- Implemented RAG architecture and vector databases for domain-specific information retrieval
- Reduced processing time by 10x through optimized data pipelines and caching strategies
- Built robust API layer with FastAPI serving multiple internal products

Max Planck Institute - Postdoctoral Fellow
Leipzig, Germany

Aug 2021 - May 2023

- Developed spectral graph theory algorithms for complex networks
- Co-led open-source software development for research community

Yahoo! Research - Research Intern
New York, NY

May 2019 - Jul 2019

- Built graph learning models on terabyte-scale social network data
- Developed distributed computing pipelines using PySpark

Wolfram Research South America - Research Programmer
Lima, Peru (Remote)

2012 - 2014

- Developed data pipelines for the Wolfram|Alpha knowledge engine
- Owned specific data domains end-to-end; worked in a globally distributed team

Open Source Projects

The Aris Program - Founder & Lead Developer

2025 - Present

aris.pub

- Open-source academic publishing platform with Python, FastAPI, Vue.js
- Real-time collaboration architecture, launching 2025

XGI (Complex Group Interactions) - Co-Lead Developer

2021 - Present

xgi.readthedocs.io

- Python library for hypergraph analysis, NumFOCUS affiliated
- NumPy pandas pytest GitHub Actions

Manim Community Edition - Core Developer

May 2020 - May 2021

manim.community

- Mathematical animation library popularized by 3Blue1Brown
- Graphics programming, code review, community management

netrd - Co-Lead Developer

Jan 2019 - Jul 2019

github.com/netsiphd/netrd

- Implemented 40+ algorithms for network reconstruction and comparison
- Led team of 12+ developers, established coding standards
- Published in Journal of Open Source Software (JOSS)

Professional Development

- Talk: "LLMs: Computation or Measurement?" - Berlin Data Meetup, 2024 (YouTube)
- Workshop: "Communicating with Data" - Designed and delivered technical workshop for enterprise clients
- Recurse Center: 12-week programming retreat focused on algorithm design and software craftsmanship (2016)

Selected Publications

- L. Torres, A. S. Blevins, D. S. Bassett, T. Eliassi-Rad. "The Why, How, and When of Representations for Complex Systems." SIAM Review 63(3), 2021.
- J. Jost, R. Mulas, L. Torres. "Spectral theory of the non-backtracking Laplacian for graphs." Discrete Mathematics 346(10), 2023.
- Co-lead developer on XGI (JOSS, 2023) and netrd (JOSS, 2021); see Open Source Projects above. Full list on Google Scholar.

Education

Ph.D. Network Science - Northeastern University, Boston, MA

2016-2021

Dissertation: Spectral Aspects of Mining Complex Networks

Advisor: Tina Eliassi-Rad

B.Sc. Mathematics - Pontificia Universidad Católica del Perú, Lima

2009-2015

Top 3% grades in 75-year history of Sciences Department

Honors Thesis: Small-World Random Graphs Beyond Watts-Strogatz